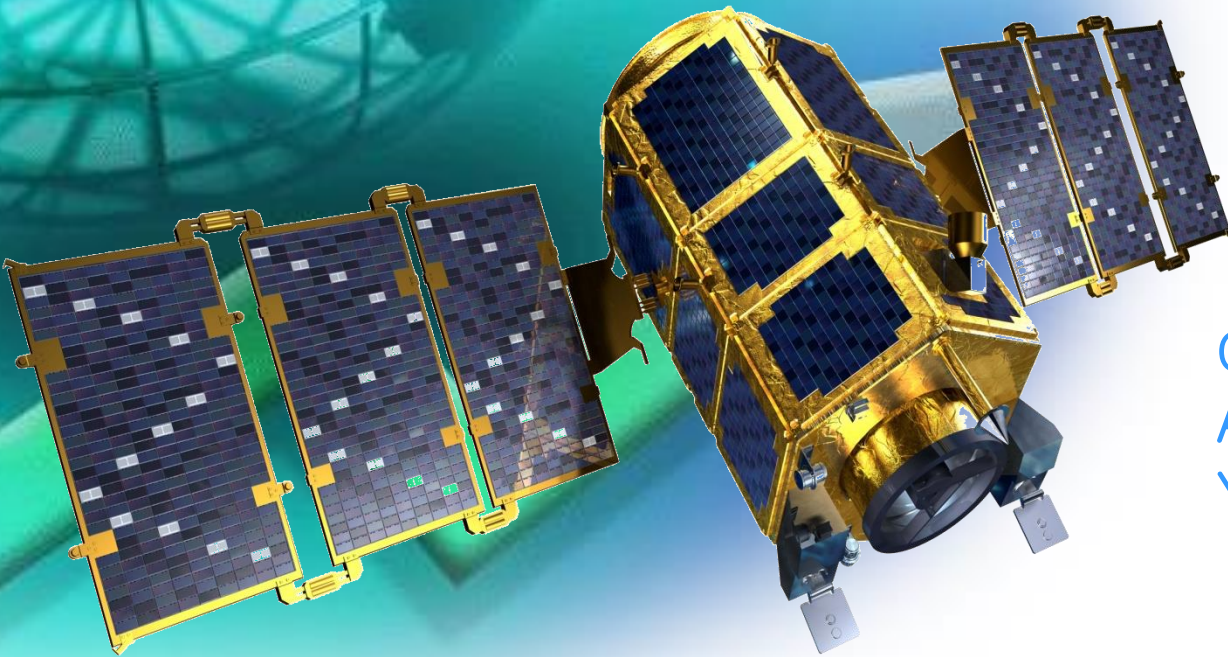
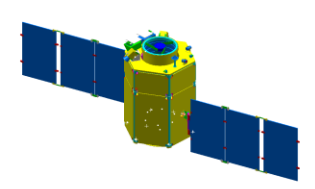


Fall 2011

Spacecraft Systems



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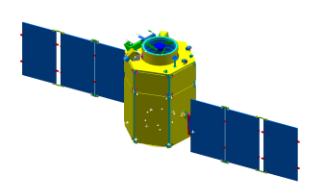


Chapter 11

SPACECRAFT SUBSYSTEMS

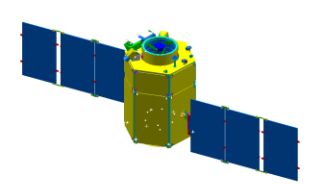


- 11.1 Attitude Determination & Control Subsystem
- 11.2 Telemetry, Tracking & Command Subsystem
- 11.3 Command & Data Handling Subsystem
- 11.4 (Electrical) Power Subsystem
- 11.5 Thermal Control Subsystem
- 11.6 Structures & Mechanism Subsystem
- 11.7 Guidance, Navigation & Control Subsystem



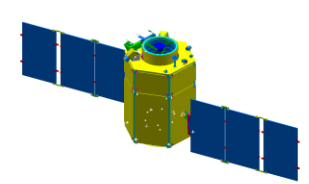
11.7 Guidance & Navigation

- ◆ Navigation and Orbit Determination (OD) mean interchangeably determining S/C Position and Velocity, or equivalently, Orbital Elements as a function of time
 - Real-Time OD
 - Provides the best estimate of where a satellite is at the present time
 - May be important for S/C and payload operations
 - Definitive OD
 - Best estimate of satellite position and orbital elements at some earlier time
 - Orbit Propagation
 - Refers to looking ahead in time from when the data was taken
 - Is used either for planning or operations



11.7 Guidance & Navigation

- ◆ Guidance and Orbit Control mean adjusting the orbit to meet some predetermined conditions
 - Orbit Maintenance
 - Refers to maintaining the orbital elements
 - Ex) Altitude Maintenance
 - Station-Keeping 
 - Refers to maintaining the satellite within a predefined box
 - Ex) Geosynchronous Station-Keeping, Constellation Maintenance.
- ◆ Navigation and Guidance function can be thought of as OD and Control Subsystem which measures and maintains the position of S/C center of mass



11.7 Guidance & Navigation

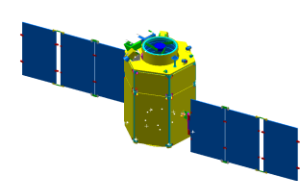
◆ 11.7.1 System Definition Process

- Traditionally, Guidance and Navigation have been a ground-operations activity
- Autonomous Navigation determines the orbit on board the satellite in real time.

TABLE 11-64. Process for Defining the Guidance and Navigation Subsystem. See text for discussion of each step.



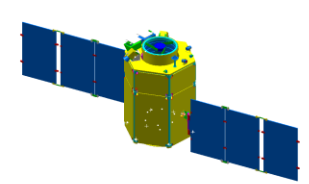
Step	Principal Issues	Where Discussed
1. Define navigation and orbit-related top-level functions and requirements	Mapping and pointing Scheduling Constellation or orbit maintenance Rendezvous or destination requirements	Secs. 1.4, 2.1, 4.2, 7.1
2. Do pointing and mapping trades to determine preliminary navigation (position) accuracy requirements	What payload functions will the navigation data be used for? Payload data processing (mapping) Payload pointing	Sec. 5.4
3. Determine whether orbit control or maintenance is needed	Geosynchronous stationkeeping Constellation stationkeeping Altitude maintenance Maintaining orbit elements Mid-course corrections	Chap. 7, Sec. 11.7.3



11.7 Guidance & Navigation

◆ 11.7.1 System Definition Process (cont'd)

4. If yes, do trade on autonomous vs. ground-based orbit control	Is reduced operations cost and risk worth introducing a nontraditional approach?	Secs. 2.1.2, 11.7.1, 11.7.3
5. Determine where navigation data is needed	Is it needed only at ground station for mission planning and data evaluation? Is it needed on board (orbit maintenance, Sun vector determination, payload pointing, target selection)? Is navigation (or target location) data needed by several end users who may get information directly from the spacecraft?	Sec. 2.1.1
6. Do autonomous vs. ground-based navigation trade	Does reduced operations cost and risk justify a nontraditional approach? Is there a need for real-time navigation data?	Secs. 2.1.1, 11.7.1
7. Select navigation method	See Sec. 11.7.2 for main options	Sec. 11.7.2
8. Define G&N system requirements	Top-level requirements should be in terms of what is needed (mapping, pointing, constellation maintenance, level of autonomy), not how the mission is done	Secs. 11.7.1, 11.7.4

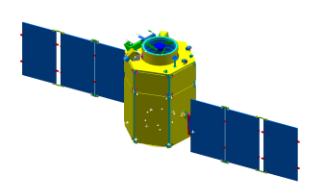


11.7 Guidance & Navigation

◆ 11.7.2 OD Systems

- Three Elements to OD Problem
 - Source and Type of Data
 - Algorithms for modeling the orbit
 - Computer Program which process the observations
- Algorithm consists of:
 - Observation Data and Model
 - Orbit Dynamics and Model
 - Estimation (or Filtering) Process
- GTDS (Goddard Trajectory Determination System)
- NORAD (North American Aerospace Defense Command)
- JPL's Deep Space Network (DSN) for interplanetary missions





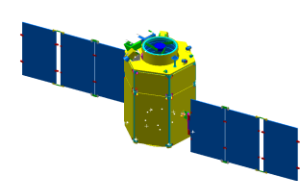
11.7 Guidance & Navigation

◆ 11.7.2 OD Systems (cont'd)

TABLE 11-65. Advantages and Disadvantages of Alternative Navigation Methods. (See also Table 11-66.)

System	Advantages	Disadvantages
<i>Ground Tracking</i>	Traditional approach Methods and tools well established	Accuracy depends on ground-station coverage Can be operations intensive
<i>TDRS Tracking</i>	Standard method for NASA spacecraft High accuracy 50m, range & range-rate data Same hardware for tracking and data links	Not autonomous Available mostly for NASA missions Requires TDRS tracking antenna

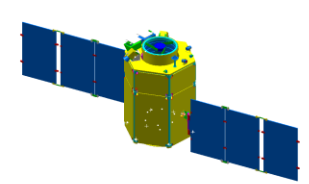
- TDRS: Tracking and Data Relay Satellite



11.7 Guidance & Navigation

◆ 11.7.2 OD Systems (cont'd)

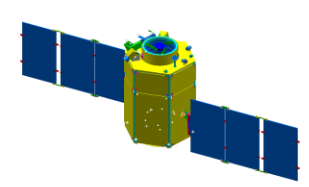
<i>Global Positioning System (GPS); GLONASS</i>	High accuracy Provides time signal as well as position	Semi-autonomous Depends on long-term maintenance and structure of GPS Orbit only (see text for discussion) Must initialize some units
<i>Microcosm Autonomous Navigation System (MANS)</i>	Fully autonomous Uses attitude-sensing hardware Provides orbit, attitude, ground look-point, and direction to Sun	First flight test in 1993 Initialization and convergence speeds depend on geometry Observations of Earth, Sun, Moon
<i>Space Sextant</i>	Could be fully autonomous Measuring the angle between a star and the limb of the moon	Flight-tested prototype only—not a current production product Relatively heavy and high power
<i>Stellar Refraction</i>	Could be fully autonomous Uses attitude-sensing hardware	Still in concept and test stage Interaction of starlight with Earth's atmosphere
<i>Landmark Tracking</i>	Can use data from observation payload sensor	Still in concept stage Landmark identification may be difficult May have geometrical singularities
<i>Satellite Crosslinks</i>	Can use crosslink hardware already on the spacecraft for other purposes	Unique to each constellation No absolute position reference Potential problems with system deployment and spacecraft failures
<i>Earth and Star Sensing</i>	Earth and stars available nearly continuously in vicinity of Earth	Cost and complexity of star sensors Potential difficulty identifying stars



11.7 Guidance & Navigation

◆ 11.7.3 Orbit Maintenance and Control

- Orbit Control is needed when:
 - Targeting to achieve an end orbit or position
 - Satellite Rendezvous
 - Interplanetary Missions
 - Overcoming secular orbit perturbations
 - Altitude Maintenance in low-earth orbit
 - Geosynchronous Station-Keeping
 - Relative Station-Keeping/Absolute Station-Keeping
 - Maintaining relative orientations
 - Constellation maintenance
- Orbit Control
 - Ground-Based Orbit Control
 - traditional method for orbit maintenance and control
 - Autonomous Orbit Control
 - Key component in a fully autonomous S/C bus
 - Reduce mission cost and risk



11.7 Guidance & Navigation

◆ 11.7.3 Orbit Maintenance and Control

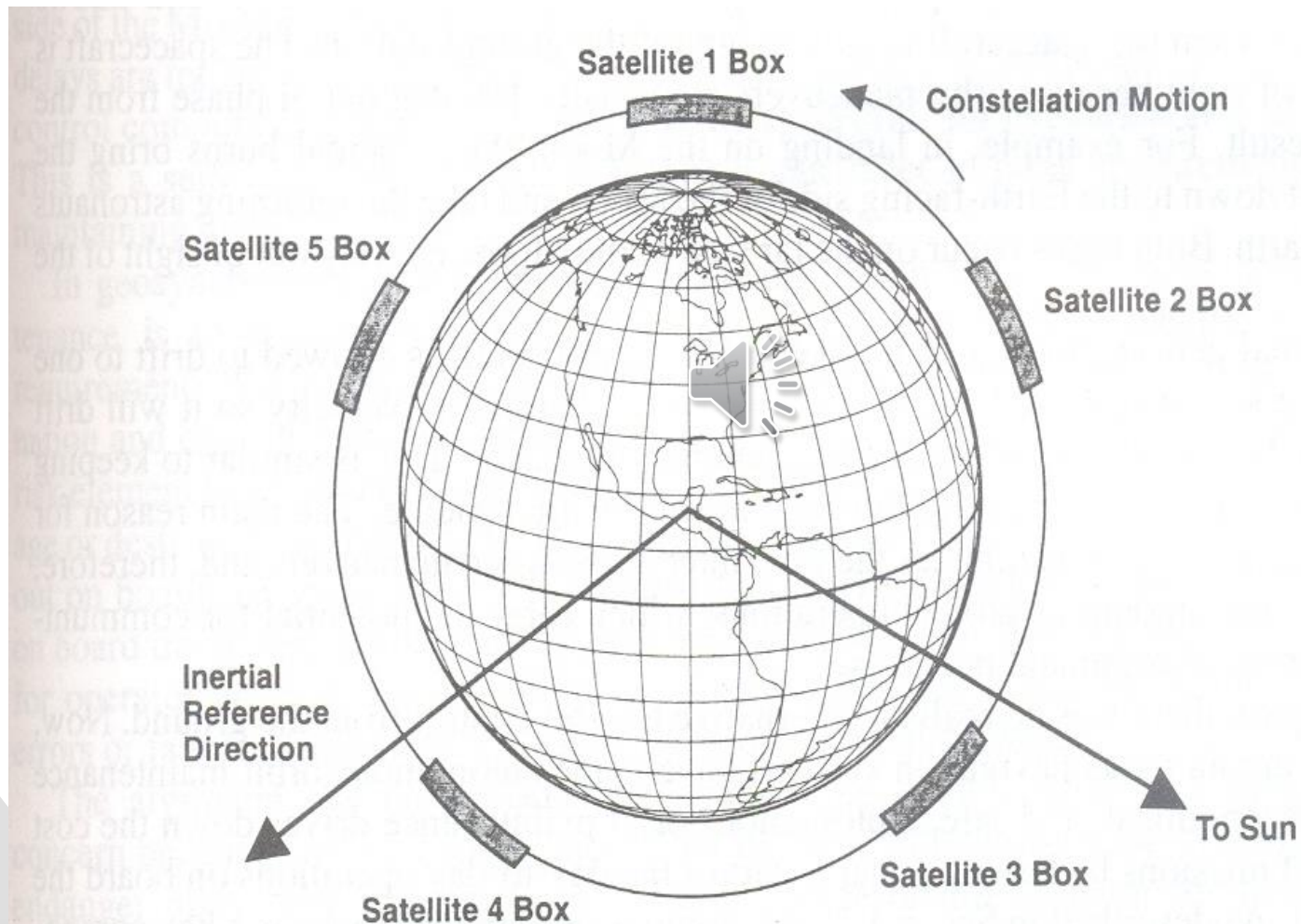


Fig. 11-51. Constellation Stationkeeping Maintains Each Satellite in a “Box” Rotating with the Constellation.